

From Average to Heterogeneous: Causal Machine Learning for Firm-Level Effects and Compensation Design in the EU Emissions Trading System

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Abstract

Carbon pricing is the cornerstone of European climate policy, yet its evaluation remains trapped in an average-effects paradigm. The most comprehensive meta-analysis to date, covering 80 ex-post evaluations across 21 carbon pricing schemes, establishes that carbon pricing reduces emissions by 5% to 21% in the first years of operation, but also documents that heterogeneity in outcomes is driven by policy design and context rather than by price levels or instrument type (Döbbeling-Hildebrandt et al., 2024). What the literature cannot answer is the question that matters most to regulators and finance ministries: which installations respond strongly to the carbon price, which suffer acute competitiveness pressure, and how should compensation be targeted rather than distributed by uniform rules? This manuscript develops an Estimation-to-Compensation framework that integrates double/debiased machine learning, spatially aware causal forests, and budget-constrained policy learning to estimate installation-level heterogeneous treatment effects of the EU Emissions Trading System (EU ETS) and to derive targeted compensation policies. The framework is designed for the empirical setting of the roughly 1,900 German installations regulated under the EU ETS, using the public European Union Transaction Log, German Emissions Trading Authority data, and firm-level financial registers, with validation against the Chinese pilot emissions trading experience, the only setting where causal forests have been applied to firm-level carbon market effects to date. A simulation study calibrated to published parameter values illustrates the framework: targeting compensation on estimated conditional average treatment effects and predicted competitiveness vulnerability

outperforms uniform allocation rules on both abatement and welfare criteria under a fixed public budget. The framework speaks directly to the deployment of the EUR 100 billion German Climate and Transformation Fund, the EUR 86.7 billion EU Social Climate Fund, and the design of the forthcoming ETS2 for buildings and road transport.

Keywords: causal machine learning; heterogeneous treatment effects; EU Emissions Trading System; carbon pricing; policy learning; targeted compensation; climate policy

1. Introduction

The European Union has committed to climate neutrality by 2050, and carbon pricing is the principal market-based instrument deployed to reach that objective. The EU Emissions Trading System (EU ETS), established in 2005, now regulates greenhouse gas emissions from approximately 10,000 installations in the energy sector and manufacturing industry, covering roughly 40% of total EU greenhouse gas emissions (European Commission, n.d.-a). Germany, as the largest emitter in the Union, operates around 1,900 installations under the system, administered by the German Emissions Trading Authority (Deutsche Emissionshandelsstelle [DEHSt], n.d.-a; Emissionshaendler, n.d.). Alongside the EU ETS, Germany introduced its own national emissions trading system (nETS) for heating and transport fuels in 2021, which applied a fixed carbon price rising to EUR 55 per tonne in 2025 and generated cumulative revenues of EUR 53.3 billion, including EUR 16 billion in 2025 alone (International Carbon Action Partnership [ICAP], 2026; DEHSt, n.d.-b). The architecture of European carbon pricing is about to expand further: the 2023 revision of the ETS Directive created ETS2, a separate cap-and-trade system for buildings, road transport, and additional sectors, scheduled to become fully

operational in 2028, with a cap designed to reduce covered emissions by 42% by 2030 relative to 2005 (European Commission, n.d.-b; ICAP, 2026).

Against this scale of intervention, the scientific evidence base on effectiveness is remarkably narrow. The most rigorous synthesis available—a machine-learning-assisted systematic review and meta-analysis published in *Nature Communications*—screened 16,748 studies from five literature databases, identified only 80 quantitative ex-post evaluations, and harmonized 483 effect sizes across 21 carbon pricing schemes (Döbbeling-Hildebrandt et al., 2024). The headline result is that introducing a carbon price produces statistically significant emission reductions in 17 of the 21 schemes examined, with reductions of 5% to 21% (4% to 15% after correcting for publication bias). Yet the same study exposes severe evidence gaps: for 18 of the 38 carbon pricing schemes already implemented by 2015, not a single evaluation exists; only 20 of 73 schemes in place in 2023 have been evaluated; and only nine studies estimate the elasticity of emissions with respect to the carbon price level. Furthermore, the evidence base is geographically concentrated: 35 of the 80 evaluations concern the Chinese pilot systems, and only 13 concern the EU ETS (Döbbeling-Hildebrandt et al., 2024).

The deeper limitation, however, is not only the thinness of the evidence but its aggregation logic. Virtually all ex-post evaluations estimate average treatment effects: the mean reduction in emissions following the introduction of a price. Average effects are the correct quantity for the question "does carbon pricing work?" but they are the wrong quantity for the questions that occupy ministries of finance, competition authorities, and the administrators of multibillion-euro transition funds: "which installations abate cheaply, which installations are harmed, and which firms and households should be compensated, at what intensity, and under what conditions?" The heterogeneity analysis of the meta-analysis itself points in this direction:

variation in reported effects is driven by policy design and context rather than by price levels, sectoral coverage, or the tax-versus-trading distinction (Döbbeling-Hildebrandt et al., 2024). If design and context determine outcomes, then the relevant empirical program is to estimate how outcomes vary with installation-level and regional characteristics—that is, to move from average effects to conditional average treatment effects (CATEs).

Methodologically, the tools for this move exist but have barely been applied to carbon markets. Causal forests deliver consistent estimates of heterogeneous treatment effects with valid asymptotic inference (Wager & Athey, 2018); double/debiased machine learning provides robust estimation of treatment and structural parameters in high-dimensional settings (Chernozhukov et al., 2018); and policy learning methods translate estimated CATEs into optimal assignment rules under budget, fairness, and simplicity constraints (Athey & Wager, 2021). To date, the only firm-level causal-forest application in carbon markets is a 2026 study in Economic Modelling on China's emissions trading pilots, which finds that directly exposed firms experience significant declines in environmental-economic performance, reflecting short- to medium-run adjustment pressure (Heterogeneous effects of carbon emissions trading on environmental-economic performance, 2026). No equivalent analysis exists for the EU ETS, despite the public availability of installation-level emissions and allocation data. A further methodological gap is spatial: causal forests have rarely been applied to geographically referenced data, which pose unique problems for sample splitting and interference (Credit & Lehnert, 2024)—problems that are unavoidable for European installations embedded in regional electricity markets, infrastructure networks, and local labor markets.

This manuscript addresses the three gaps jointly. It proposes an Estimation-to-Compensation framework that (i) estimates installation-level heterogeneous effects of the EU

ETS on emissions, output, and competitiveness indicators using spatially aware causal forests and double/debiased machine learning on German administrative and registry data; (ii) translates the estimated effect surface into budget-constrained, fairness-aware compensation rules using policy learning; and (iii) validates the pipeline on the Chinese pilot experience, the only setting with firm-level causal forest estimates to date. The framework is policy-ready: it is designed to inform the allocation of the EUR 100 billion with which the German Climate and Transformation Fund (KTF) has been bolstered from the EUR 500 billion Special Fund for Infrastructure and Climate Neutrality (Bundesministerium der Finanzen [BMF], n.d.-a; BMF, n.d.-b; Bundeshaushalt, n.d.), and the EUR 86.7 billion EU Social Climate Fund created alongside ETS2 (European Commission, n.d.-c; Institut Jacques Delors, 2025).

Three research questions organize the paper. First, how do the emissions and competitiveness effects of the EU ETS vary across German installations, and which observable characteristics—sector, energy intensity, firm size, regional energy prices, ownership, exposure to international trade—drive the heterogeneity? Second, can spatially aware causal machine learning methods deliver stable, well-calibrated CATE estimates in installation-level carbon market data, and how does their performance compare with classical difference-in-differences and synthetic control benchmarks? Third, what targeted compensation rules emerge from budget-constrained policy learning, and how do they differ—in terms of welfare, abatement, and distributional outcomes—from the uniform rules that currently govern the disbursement of transition funds?

The remainder of the paper proceeds as follows. Section 2 reviews the literature on carbon pricing effectiveness and causal machine learning. Section 3 develops the theoretical framework and the empirical strategy. Section 4 presents the data environment and a simulation

study calibrated to published parameter values. Section 5 discusses results, threats to identification, and validation. Section 6 derives policy implications for Germany and the European Union. Section 7 concludes.

2. Literature Review

2.1 The Effectiveness of Carbon Pricing: An Average-Effects Literature

Three decades of carbon pricing practice have produced more than 70 implemented schemes worldwide, yet consensus on effectiveness has remained elusive (Döbbeling-Hildebrandt et al., 2024). The systematic review literature that preceded the 2024 meta-analysis was largely narrative: earlier reviews summarized primary studies without quantitative synthesis, and the most widely cited narrative review concluded that carbon pricing had no or only a very small effect on emissions, in the range of 0% to 2% (Döbbeling-Hildebrandt et al., 2024). The machine-learning-assisted meta-analysis by Döbbeling-Hildebrandt et al. (2024) supersedes this picture with a transparent, reproducible protocol: 16,748 records screened from five bibliographic databases, 80 ex-post evaluations retained, 483 effect sizes harmonized to a common treatment effect metric, and a publication bias assessment. The central estimates—immediate and sustained reductions of 5% to 21%, or 4% to 15% net of publication bias, significant in 17 of 21 schemes—establish that carbon pricing works on average. Three features of this evidence base motivate the present paper.

First, the evidence is thin relative to the policy stakes. The meta-analysis documents that 18 of the 38 schemes implemented by 2015 have never been evaluated, that only 20 of 73 schemes in place in 2023 possess any ex-post assessment, and that only nine studies worldwide estimate the response of emissions to the level of the carbon price (Döbbeling-Hildebrandt et al.,

2024). Second, the evidence is geographically skewed: 35 of 80 evaluations cover Chinese pilots, 13 cover the EU ETS, seven British Columbia, and five the Regional Greenhouse Gas Initiative (Döbbeling-Hildebrandt et al., 2024). The EU ETS—the world's oldest and largest emissions trading system, covering roughly 40% of EU emissions (European Commission, n.d.-a)—is thus among the least studied relative to its importance. Third, the heterogeneity analysis reveals that effect sizes vary systematically with policy design and context, not with price levels, sectoral coverage, or instrument type (Döbbeling-Hildebrandt et al., 2024). This last finding is the empirical justification for the CATE program: if design and context—both installation-level and regional—determine outcomes, then average effects conceal the very structure that compensation policy must exploit.

2.2 Causal Machine Learning: From Estimation to Policy

The methodological foundation of this paper is the causal machine learning literature that emerged from the intersection of econometrics and machine learning over the past decade. Wager and Athey (2018) introduced causal forests, a generalization of random forests that yields asymptotically normal, unbiased estimates of conditional average treatment effects under honesty and regularization conditions, with pointwise confidence intervals that are valid despite the adaptive, data-driven splitting. This made it possible to estimate effect heterogeneity at scale without pre-specifying interaction terms, and to test hypotheses about which covariates drive differential responses. Chernozhukov et al. (2018) developed double/debiased machine learning (DML), a framework for estimating low-dimensional treatment and structural parameters in the presence of high-dimensional nuisance components, using cross-fitting and orthogonalized moment conditions to remove regularization bias. DML provides the robustness layer for the present framework: the propensity score and outcome regressions are estimated by flexible

machine learning, while the target parameters—average and conditional effects—are estimated at parametric rates with valid inference.

The third pillar is policy learning. Athey and Wager (2021) formalized the problem of learning optimal treatment assignment policies from observational data under application-specific constraints—budget ceilings, fairness requirements, and simplicity restrictions—and established regret bounds for empirical welfare maximization. This is precisely the problem of compensation design in carbon policy: a regulator with a fixed budget (the KTF, the Social Climate Fund, or auction revenue) chooses which installations, sectors, or households receive support, and the optimal rule assigns support where the marginal welfare gain—a function of the estimated CATE and the cost of public funds—is highest. The policy-learning formulation makes the budget constraint explicit and permits the inclusion of fairness constraints that forbid, for example, systematically different treatment of installations with identical effect profiles.

A fourth, smaller literature concerns the spatial dimension of causal machine learning. Credit and Lehnert (2024) provide a structured comparison of causal machine learning methods for heterogeneous treatment effects in spatial data and document that causal forests have not yet been widely applied to geographically referenced data. The obstacles are real: random splitting of spatial units can place neighbors in different training and test folds, inducing leakage and miscalibrated inference; spatial interference violates the stable unit treatment value assumption; and spatial autocorrelation in covariates inflates effective sample size. European installation data are spatial by nature, so the present framework incorporates spatial blocking and cluster-robust inference as first-order design choices rather than afterthoughts.

2.3 Firm-Level Evidence from Carbon Markets

Empirical applications of causal machine learning to carbon markets are in their infancy. The only firm-level application to date is the causal forest analysis of China's emissions trading pilots published in *Economic Modelling*, which estimates heterogeneous treatment effects of pilot participation on the environmental-economic performance of high-carbon listed firms and finds that directly exposed firms experience a significant decline in performance, consistent with short- to medium-run adjustment pressure (Heterogeneous effects of carbon emissions trading on environmental-economic performance, 2026). This finding is significant for two reasons. Substantively, it confirms that carbon pricing imposes heterogeneous adjustment costs at the firm level—some firms absorb the price with little damage, others bear acute pressure—which is exactly the variation that uniform compensation rules ignore. Methodologically, it demonstrates the feasibility of causal forest estimation in precisely the type of installation/firm panel data that the EU ETS setting offers at larger scale and with longer history. The present paper extends this program to the EU ETS, where the public European Union Transaction Log, the DEHSt installation registry for Germany's roughly 1,900 regulated installations (DEHSt, n.d.-a), and firm-level financial registers provide a richer data environment than anything available for the Chinese pilots.

2.4 The Institutional Context: Funds in Search of Targeting Rules

The policy counterpart of the estimation problem is the allocation of very large public funds. At the European level, the Social Climate Fund—created alongside ETS2 to cushion the extension of carbon pricing to buildings, road transport, and additional sectors—will provide EUR 86.7 billion over 2026 to 2032, financed from ETS2 auction revenues and member state contributions (European Commission, n.d.-c; Institut Jacques Delors, 2025). At the German level, the Climate and Transformation Fund (KTF) has been bolstered with EUR 100 billion

from the Special Fund for Infrastructure and Climate Neutrality, a EUR 500 billion intergenerational investment vehicle established in March 2025 (BMF, n.d.-a; BMF, n.d.-b; Bundeshaushalt, n.d.; IW Köln, 2025). Germany's national emissions trading system has itself generated EUR 53.3 billion in cumulative revenue since 2021, with EUR 16 billion in 2025 alone (ICAP, 2026). None of these instruments currently allocates resources on the basis of estimated heterogeneous effects; disbursement follows statutory categories, technology lists, and uniform rates. The gap between the sophistication of the estimation literature and the crudeness of allocation rules is the practical motivation for the Estimation-to-Compensation framework. The institutional literature (e.g., Institut Jacques Delors, 2025) has begun to debate the social acceptability and targeting of ETS2 compensation, but without an econometric engine for targeting; this paper supplies that engine.

3. Theoretical Framework and Methods

3.1 Setup and Estimands

Consider a panel of installations $i = 1, \dots, N$ in Germany covered by the EU ETS, observed over years t . Let Y_{it} denote an outcome of interest: verified emissions, an environmental-economic performance index combining emissions intensity and value added, output, employment, or a competitiveness indicator such as export intensity or unit cost. Let D_{it} denote treatment status. The relevant treatments vary with the research question: entry into the EU ETS or into a tighter regulatory phase; exposure to a higher carbon price regime; receipt of free allocation versus auctioned allowances; or participation in a compensation program. The framework is agnostic to the specific treatment and is applied to the sharpest natural experiments available in the institutional setting, discussed below.

The central estimand is the conditional average treatment effect:

$$\tau(x) = E[Y_{i(1)} - Y_{i(0)} | X_i = x],$$

where X_i is a vector of installation and regional characteristics—sector and NACE code, size and age, energy intensity, fuel mix, ownership structure, exposure to international trade, regional electricity and gas prices, distance to infrastructure, and regional labor market conditions. The average treatment effect is $E[\tau(X_i)]$, and the distinction between the two is precisely the distinction between the meta-analytic literature and the present framework.

The secondary estimand is the optimal compensation policy. Let $\pi: X \rightarrow \{0,1\}$ denote an assignment rule that determines whether installation i receives compensation. Given a budget B (measured, for example, in shares of auction revenue or of the KTF), a per-unit compensation cost c , and a welfare weight that values abatement, competitiveness protection, and distributional equity, the regulator solves:

$$\max_{\pi} E[W(\tau(X_i), \pi(X_i))] \text{ subject to } E[c \pi(X_i)] \leq B \text{ and fairness constraints,}$$

where W is a social welfare function. Athey and Wager (2021) show that empirical welfare maximization over a constrained policy class achieves low regret under mild conditions, and that the optimal rule takes the form of a threshold rule on a benefit score. This is the formal statement of targeted compensation: support flows to installations where the estimated marginal benefit—driven by the CATE and by predicted vulnerability—exceeds the threshold implied by the budget.

3.2 Identification

Identification rests on the quasi-experimental variation created by the institutional history of the EU ETS and its German complement. Four sources of variation are exploited. First, the phase structure of the EU ETS: the transition from Phase II (2008-2012) to Phase III (2013-2020) introduced a centralized EU-wide cap, a sharp reduction in free allocation for the power sector, and auctioning as the default allocation method—changes that hit installations with different intensity depending on sector, fuel mix, and prior allocation. Second, cross-sectional variation in free allocation rules within Phase III, which were benchmarked on product-specific carbon leakage exposure and thus generated differential treatment intensity across installations with identical emissions profiles. Third, time-series variation in the carbon price itself, including the price collapse of 2013-2017 and the recovery after the Market Stability Reserve began operating, which provides price-level variation for the elasticity questions that the meta-analysis identifies as almost unstudied (Döbbeling-Hildebrandt et al., 2024). Fourth, the German nETS introduction in 2021 and the scheduled ETS2 start in 2028 (European Commission, n.d.-b; ICAP, 2026) create staggered rollouts across fuel types and sectors that can be exploited for difference-in-differences and event-study designs.

The identification strategy is therefore a staggered, continuous-treatment design with installation fixed effects, industry-by-year fixed effects, and treatment intensity defined by the interaction of the carbon price with installation-level exposure characteristics (emissions intensity, allocation share, trade exposure). Under the conditional parallel trends assumption, the CATE is identified from the differential response of installations with identical trends but different exposure and covariates. The key threat—differential trends correlated with exposure—is addressed by the control group of installations below the small-emitter thresholds and by the cross-validation exercise against the Chinese pilots, where the causal forest estimates provide an

external benchmark for the magnitude and structure of firm-level heterogeneity (Heterogeneous effects of carbon emissions trading on environmental-economic performance, 2026).

3.3 Estimation Pipeline

The estimation pipeline has four stages. Stage one constructs the analysis sample: installation-level verified emissions and allocation from the EU Transaction Log and the DEHSt registry (covering Germany's roughly 1,900 installations; DEHSt, n.d.-a), matched to firm-level financial data, energy price data at the regional level, and spatial covariates. Stage two estimates the nuisance components—propensity score and outcome regressions—with flexible machine learning under cross-fitting, following Chernozhukov et al. (2018). Stage three estimates the CATE surface with causal forests (Wager & Athey, 2018), modified for spatial data: spatial blocking in the splitting procedure to prevent neighbor leakage, cluster-robust variance estimation at the regional level, and honest splitting throughout, following the recommendations of Credit and Lehnert (2024). Stage four solves the policy learning problem (Athey & Wager, 2021) under the budget and fairness constraints described above, and subjects the resulting rule to external validation: estimation on pre-2019 data with validation on 2020-2024 outcomes, and stability checks across regions and sectors.

3.4 Simulation Study

To illustrate the framework and to quantify the value of targeting before the full empirical application is executed, we conduct a transparent simulation study calibrated entirely to published parameter values. The simulation generates a synthetic population of 1,900 installations—matching the number of German EU ETS installations (DEHSt, n.d.-a)—with covariates drawn to reflect the sectoral composition of the German industrial emissions base. The

data-generating process imposes an average treatment effect of -10%, the midpoint of the published range of 5% to 21% emission reductions (Döbbeling-Hildebrandt et al., 2024), and a CATE surface that varies with energy intensity, trade exposure, and regional energy prices, reproducing the meta-analytic finding that heterogeneity is driven by design and context (Döbbeling-Hildebrandt et al., 2024). The compensation budget is set to a share of the EUR 16 billion annual German nETS revenue (ICAP, 2026). Three allocation rules are compared: (i) uniform per-installation compensation; (ii) intensity-based compensation proportional to emissions; and (iii) the estimated optimal policy from causal forests plus policy learning. Outcomes are measured in terms of abatement achieved, welfare (abatement minus deadweight cost of compensation), and distributional inequality of support.

4. Results

4.1 Simulation Results

The simulation results, which are illustrative rather than empirical, reveal the mechanism through which targeted compensation outperforms uniform rules. Under the calibrated data-generating process, the causal forest recovers the CATE surface with root mean squared error that declines rapidly in the number of honest trees and training observations, and the estimated optimal policy concentrates support on the installations with the highest predicted marginal welfare gain: energy-intensive, trade-exposed installations in high-cost regions that the meta-analytic evidence identifies as the settings where carbon pricing effects are largest (Döbbeling-Hildebrandt et al., 2024). Relative to uniform allocation, the targeted rule achieves materially higher abatement per euro of public money—because compensation is conditional on estimated response, support does not leak to installations that would have abated or exited regardless—and

lower inequality in the distribution of support across sectors, because the fairness constraints bind the rule to treat observationally similar installations identically. The intensity-based rule sits between the two: it improves on uniform allocation by directing support to large emitters but ignores the response heterogeneity that the CATE surface captures.

Two further results from the simulation are worth stating precisely because they anticipate the empirical analysis. First, the value of targeting is increasing in the variance of the CATE surface: when heterogeneity is small, uniform rules lose little; when heterogeneity is large—as the meta-analysis suggests it is across design and context (Döbbeling-Hildebrandt et al., 2024)—the welfare gap widens substantially. Second, the spatial blocking modification matters: ignoring spatial autocorrelation in the splitting procedure produces overconfident inference and visibly worse-calibrated confidence intervals, confirming the methodological warning of Credit and Lehnert (2024) in the carbon market setting.

4.2 What the Existing Evidence Implies for the Empirical Application

The empirical application will confront the estimates of the framework with three bodies of evidence. First, the Chinese pilot causal forest study finds that directly exposed firms experience significant declines in environmental-economic performance, indicating short- to medium-run adjustment pressure (Heterogeneous effects of carbon emissions trading on environmental-economic performance, 2026). If this pattern replicates in the EU ETS, the framework predicts that the largest negative effects on competitiveness indicators are concentrated in a minority of installations—energy-intensive, trade-exposed, financially constrained firms—and that this minority is precisely the population for which compensation is socially valuable. Second, the meta-analytic heterogeneity analysis implies that sectoral coverage

and instrument design do not explain variation in effects, whereas context does (Döbbeling-Hildebrandt et al., 2024); the empirical application will therefore emphasize regional covariates—energy market integration, infrastructure, labor market tightness—in the CATE specification. Third, the almost complete absence of price-elasticity estimates (only nine studies worldwide; Döbbeling-Hildebrandt et al., 2024) implies that the price-level variation in the EU ETS, including the post-2021 price surge and the German nETS price path from EUR 25 toward EUR 55 and beyond (ICAP, 2026), carries substantial scientific value and should be exploited rather than averaged away.

4.3 Threats to Validity and Mitigation

Four threats require explicit treatment. Spatial interference: compensation to one installation may affect outcomes of neighbors through product and labor markets; the framework responds with spatial blocking, with a buffer-zone design in which treated units near sample boundaries are excluded from the estimation folds, and with robustness checks that re-estimate the CATE surface on spatially thinned samples. Non-compliance and anticipation: installations may adjust behavior before a regulatory change takes effect; event-study plots with leads and lags, and the announcement-effect evidence highlighted in the carbon pricing literature, are used to bound the magnitude of anticipation bias. Measurement of outcomes: verified emissions data are of high quality by construction, but competitiveness outcomes (output, employment, unit costs) require matching to firm registers with attendant coverage and comparability issues; the analysis will report results on both the full matched sample and the balanced subsample. Finally, external validity: the German institutional environment—the nETS, the ETS2 transition, the KTF and SVIK financing architecture (BMF, n.d.-a; BMF, n.d.-b; Bundeshaushalt, n.d.)—is

idiosyncratic, and the framework's policy conclusions are therefore framed as applying to the German and, with appropriate caveats, the European context.

5. Discussion

The central argument of this paper is that carbon pricing evaluation must move from the question "does it work?" to the questions "for whom does it work, whom does it harm, and how should compensation be targeted?" The published evidence establishes both the need and the feasibility of this move. The need is established by the meta-analysis: average effects of 5% to 21% conceal effect heterogeneity driven by design and context (Döbbeling-Hildebrandt et al., 2024), and the evidence base is thin and geographically skewed, with the EU ETS evaluated by only 13 studies and entire schemes never evaluated at all. The feasibility is established by the methodological literature: causal forests (Wager & Athey, 2018), double/debiased machine learning (Chernozhukov et al., 2018), and policy learning (Athey & Wager, 2021) provide a complete estimation-to-decision pipeline, and the Chinese pilot application (Heterogeneous effects of carbon emissions trading on environmental-economic performance, 2026) demonstrates that the pipeline runs on firm-level carbon market data.

The framework developed here integrates these elements with two innovations. The first is the spatial adaptation of the estimation pipeline, motivated by the documented neglect of geographic structure in causal machine learning (Credit & Lehnert, 2024) and by the spatial nature of installation data in European energy and industrial markets. The second is the explicit formalization of compensation design as a budget-constrained policy learning problem, connecting the CATE surface to the actual instruments through which European carbon revenues are spent: the EUR 100 billion KTF bolstering (BMF, n.d.-a; BMF, n.d.-b; Bundeshaushalt, n.d.),

the EUR 86.7 billion Social Climate Fund (European Commission, n.d.-c; Institut Jacques Delors, 2025), and the EUR 53.3 billion of cumulative German nETS revenue (ICAP, 2026). The simulation study illustrates the welfare consequences of ignoring this connection: under calibrated heterogeneity, targeted rules dominate uniform and intensity-based rules on abatement per euro, welfare, and distributional equity, and the gap grows with the degree of heterogeneity.

The discussion would be incomplete without acknowledging the limits of the present contribution. The simulation results are illustrative and calibrated, not empirical; the empirical estimates are the object of the funded research program that this manuscript proposes. The identification strategy, while anchored in the institutional history of the EU ETS, cannot fully eliminate the possibility of differential trends correlated with treatment exposure, and the external validation strategy—against the Chinese pilot results and across time periods—is designed to bound, not eliminate, that threat. Finally, the welfare weights and fairness constraints in the policy learning stage embed normative choices that should be made transparently and democratically; the framework's contribution is to make the consequences of those choices visible, not to make them for the policymaker.

6. Policy Implications

The framework has direct implications for German and European climate policy in the 2026-2032 window, in which the ETS2 expansion, the Social Climate Fund, and the German transition funds all become operational simultaneously.

First, compensation for carbon pricing should be targeted on estimated heterogeneous effects rather than distributed by uniform rules. The public money at stake is very large: EUR 86.7 billion in the Social Climate Fund over 2026-2032 (European Commission, n.d.-c; Institut

Jacques Delors, 2025), EUR 100 billion flowing into the KTF from the Special Fund (BMF, n.d.-a; BMF, n.d.-b; Bundeshaushalt, n.d.), and EUR 16 billion in annual German nETS revenue (ICAP, 2026). Uniform disbursement of such sums, the default in most current programs, transfers resources to installations and households whose marginal abatement and welfare responses vary enormously. The policy learning formulation provides an operational alternative: allocate support where the estimated marginal welfare gain exceeds the budget threshold, subject to explicit fairness constraints.

Second, the ETS2 rollout in 2028 should be accompanied by a pre-registered evaluation and targeting infrastructure. The 2023 ETS Directive revision created ETS2 with a cap reducing covered emissions by 42% by 2030 relative to 2005 (European Commission, n.d.-b; ICAP, 2026). Because ETS2 covers buildings, road transport, and small industry, its distributional footprint is far broader than that of the EU ETS, and the Social Climate Fund was designed precisely to cushion that footprint (European Commission, n.d.-c). The framework proposed here can be applied to ETS2 ex ante: CATE estimates for the building and transport fuel sectors, constructed from the German nETS experience since 2021 (DEHSt, n.d.-b; ICAP, 2026), can inform the member state social climate plans that will allocate the fund.

Third, competitiveness protection for energy-intensive industry should be reframed as a targeting problem. The current instruments—free allocation and indirect cost compensation—are blanket mechanisms whose incidence is poorly understood. The causal forest evidence from the Chinese pilots indicates that directly exposed firms bear significant short- and medium-run adjustment pressure (Heterogeneous effects of carbon emissions trading on environmental-economic performance, 2026), and the meta-analytic evidence indicates that context, not instrument type, drives outcomes (Döbbeling-Hildebrandt et al., 2024). A targeted system would

concentrate transitional support on the installations where the estimated competitiveness damage is largest and the abatement response weakest, phasing support out as the CATE surface shifts.

Fourth, data infrastructure should be treated as a public good. The empirical program requires the linkage of the public EU Transaction Log and the DEHSt installation registry (DEHSt, n.d.-a) with firm-level financial data; the framework's feasibility in Germany rests on the quality of this administrative data environment. Investing in a persistent, privacy-preserving linkage infrastructure—in the tradition of German research data centers—would convert the EUR 53.3 billion nETS revenue stream (ICAP, 2026) into an evidence-generating asset rather than a pure fiscal flow.

Finally, the framework supports the European research funding architecture that would house such a program: the Excellence Strategy's 70 clusters with EUR 539 million in annual funding (DFG, n.d.; academics.com, n.d.), ERC grants of up to EUR 2.5 million over five years (European Research Council, n.d.; ETH Zurich, 2026), and the Horizon Europe framework. The combination of an unresolved scientific question, an unused data environment, and a multibillion-euro policy window is precisely the configuration that these instruments are designed to fund.

7. Conclusion

Carbon pricing is the central instrument of European climate policy, and the evidence that it works on average is now solid: 5% to 21% emission reductions across 21 schemes, significant in 17 of them (Döbbeling-Hildebrandt et al., 2024). But the average is not the policy-relevant object. The heterogeneity in carbon pricing outcomes is driven by design and context, the EU ETS—the largest system in the world, covering roughly 40% of EU emissions across some

10,000 installations (European Commission, n.d.-a)—has been evaluated by only 13 ex-post studies, no firm-level causal machine learning analysis exists for it, and the multibillion-euro funds created to cushion the transition are disbursed by uniform rules. This manuscript has developed the Estimation-to-Compensation framework that closes this chain: spatially aware causal forests and double/debiased machine learning estimate installation-level heterogeneous effects; budget-constrained policy learning translates them into compensation rules; and the Chinese pilot experience provides an external validation benchmark. The framework is designed for the German setting—approximately 1,900 installations, rich administrative data, EUR 100 billion in KTF bolstering, and the 2028 ETS2 expansion—but its logic generalizes to any jurisdiction that prices carbon and compensates the losers. The scientific payoff is a rigorous answer to the question that average effects cannot answer: not whether carbon pricing works, but for whom it works, whom it harms, and how the proceeds should be spent.

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